

3.1 Motion of a projectile

Further Mechanics · 2026 examination · Concept guide and original exam-style practice

Understand the concept

Resolve launch velocity into perpendicular components. Horizontal acceleration is zero and vertical acceleration is $-g$ when air resistance is ignored. Use common time to combine components; at maximum height vertical velocity is zero. Range at equal launch/landing height is $u^2 \sin(2\theta)/g$.

Worked example: follow the reasoning

With $u=20 \text{ m s}^{-1}$ at 30° , vertical launch speed is 10 m s^{-1} and time to apex $\approx 1.02 \text{ s}$.

Structured practice

Answer in the space provided. Show working and units where relevant.

1 A particle is projected at 25 m s^{-1} at 30° above horizontal. Find maximum height, [3]
 $g=10 \text{ m s}^{-2}$.

Final answer: _____

Answers and explanation

Question 1 [3 marks]

$u_y = 25 \sin 30^\circ = 12.5$ [1]; at top $v_y = 0$, $h = u_y^2 / (2g)$ [1] = 7.8125 m [1].

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or calculation error; then retry without looking at the solution.

Syllabus: Further Mathematics 9231, 2026–2027.

<https://www.cambridgeinternational.org/Images/697357-2026-2027-syllabus.pdf>

Original FagusED teaching material. Question layout and mark presentation informed by the May/June 2026 papers supplied by the user. The practice question is newly written and is not a Cambridge past-paper question.